05: Conditions

Stat 230 - Spring 2026

> **Note**
>
> This is the quarto version of the R activity, designed for opening in maize/RStudio. If you prefer to use the tutorial version, you can use https://aluby.github.io/stat230/activities/05-conditions-tutorial

```
library(tidyverse)
library(ggformula)
theme_set(theme_minimal())
```

1 Example 1: Making residual plots in R

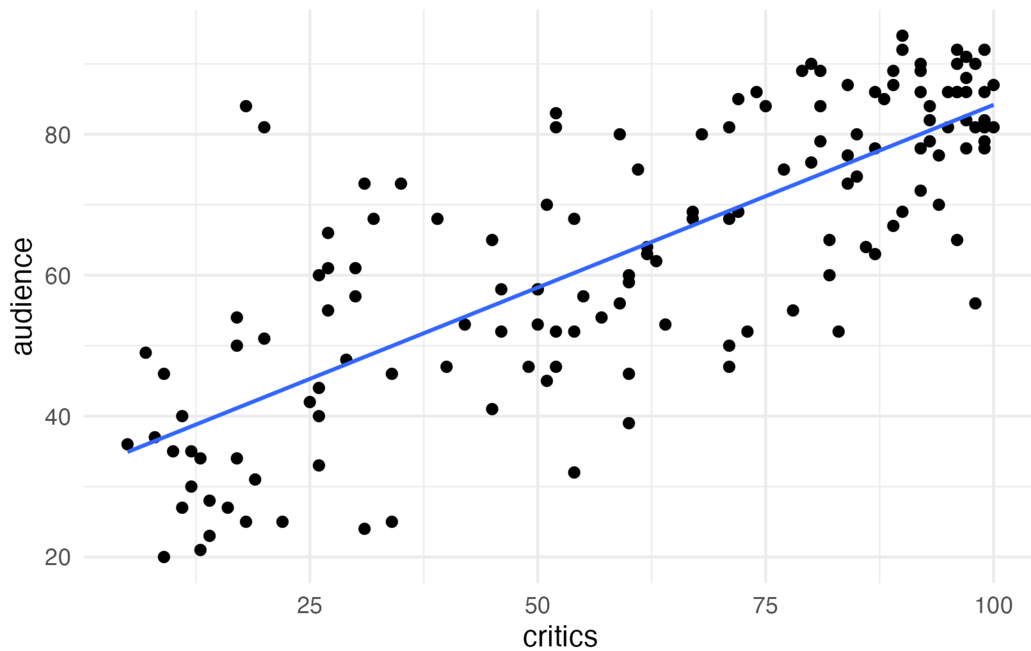
Recall the `movies` dataset from the R activity we did on Day 02. (head() prints out the first few rows)

```
movies <- read.csv("https://aloy.github.io/stat230-materials/data/movie_scores.csv")
head(movies)
```

```
                     film year critics audience
1 Avengers: Age of Ultron 2015      74       86
2              Cinderella 2015      85       80
3                 Ant-Man 2015      80       90
4        Do You Believe? 2015      18       84
5  Hot Tub Time Machine 2 2015      14       28
6       The Water Diviner 2015      63       62
```

On that day, we made the following scatterplot:

```
gf_point(audience ~ critics, data = movies) |>
  gf_lm()
```



and fit this regression model:

```
movie_lm <- lm(audience ~ critics, data = movies)
movie_lm
```

```

Call:
lm(formula = audience ~ critics, data = movies)

Coefficients:
(Intercept)      critics
    32.3155       0.5187
```

1.1 Finding residuals

In order to make residual plots, we first need to find the residuals. While we can do this by hand for a couple of points, it would be extremely tedious to do it for *every single data point*! Instead, we'll use the `augment()` function from the {broom} R package to created an augmented movie dataset. This takes the original X and Y variables and adds the $\hat{y}$ values (`.fitted`) and residuals (`.resid`) (along with some other columns that we'll use later)

Run the following code chunk and take a peek at the `movie_aug` dataset. Can you find the:

- original X variable?
- original Y variable?
- Predictions?
- Residuals?

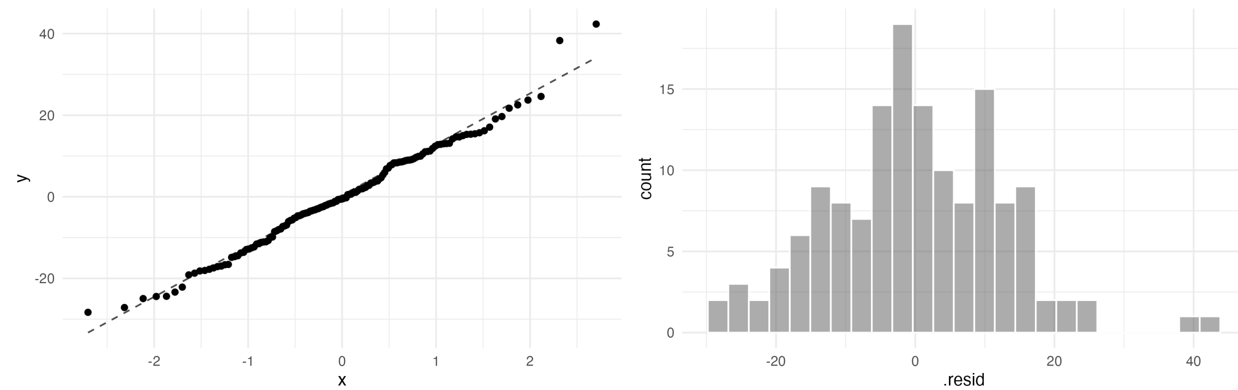
```
library(broom)
movie_aug <- augment(movie_lm)
movie_aug
```

```
# A tibble: 146 x 8
   audience critics .fitted .resid     .hat .sigma    .cooksd .std.resid
      <int>   <int>   <dbl>  <dbl>    <dbl>  <dbl>      <dbl>      <dbl>
 1       86      74    70.7  15.3  0.00816    12.5 0.00618         1.23
 2       80      85    76.4   3.60 0.0113     12.6 0.000474        0.289
 3       90      80    73.8  16.2  0.00963    12.5 0.00818         1.30
 4       84      18    41.7  42.3  0.0208     12.1 0.123           3.41
 5       28      14    39.6 -11.6  0.0235     12.5 0.0105         -0.934
 6       62      63    65.0  -2.99 0.00688    12.6 0.000199       -0.239
 7       53      42    54.1  -1.10 0.00954    12.6 0.0000374      -0.0882
 8       64      86    76.9 -12.9  0.0116     12.5 0.00633        -1.04
 9       82      99    83.7  -1.66 0.0179     12.6 0.000163       -0.134
10       87      89    78.5   8.52 0.0129     12.6 0.00305         0.684
# i 136 more rows
```

1.2 Making residual plots

The code below makes a normal QQ plot and a histogram of the residuals. Do you think the normality assumption is satisfied?

```
gf_qq(~.resid, data = movie_aug) |>
  gf_qqline()
gf_histogram(~.resid, data = movie_aug, col = "white")
```



1.2.1 Checkpoint question

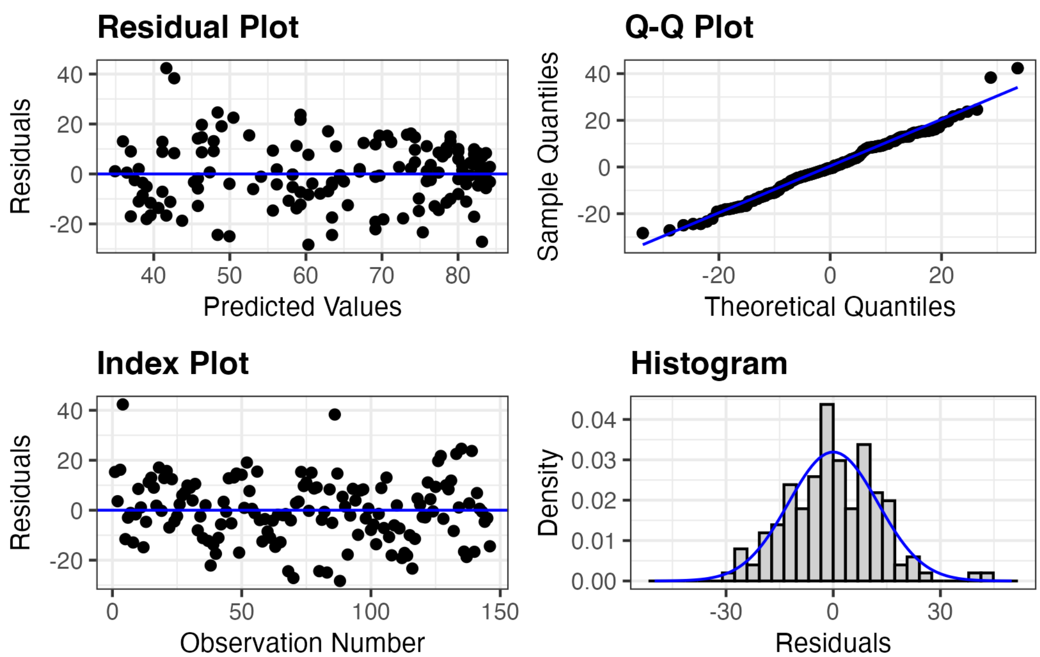
Use the `movie_aug` dataset to make a scatterplot with `.fitted` on the x-axis and `.resid` on the y-axis. List the SLR conditions you can assess with this plot, and whether or not they appear to be satisfied.

```
# your code here
```

1.3 A shortcut

We will make a **ton** of residual plots this term. A shortcut is to use the `resid_panel` function within the {ggResidpanel} package:

```
library(ggResidpanel)
resid_panel(movie_lm)
```



1.3.1 Checkpoint question

Which panel(s) would you look at for each of the following SLR conditions. List as many panels as possible, along with how you would identify any violations:

- Linearity:
- Independence:
- Normality:
- Equal variance:

2 Example 2: R^2 in R

2.0.1 Checkpoint question

Find the R^2 value for the `movie_lm` model using `summary()`. Confirm that it is equal to r^2

```
# your code here
```

2.1 Summary

2.1.1 Checkpoint question

Suppose that we fit a linear regression model to a sample of 1,000 observations and we find evidence in the normal Q-Q plot of a violation of the normality assumption, but no evidence in the residual plot of violations of the linearity and constant variance assumptions and no reason to suspect a violation of the independence assumption. Which of the following inferences are valid, and which are not?:

- p-value from the hypothesis test $H_0 : \beta_1 = 0$ vs $H_A : \beta_1 \neq 0$
- Confidence interval for β_1
- Confidence interval for the mean of Y at some X value
- Prediction interval for Y at some X value

2.2 Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
`augment(model)`	Create augmented dataset that includes predictions and residuals
`gf_qq(~variable, data)`	Make a qq plot
`gf_qq_line()`	Overlay a straight line on the qq plot
`resid_panel(model)`	Shortcut for making a panel of residual plots